



UNIVERSITAT DE
BARCELONA

Ana Cano Herranz

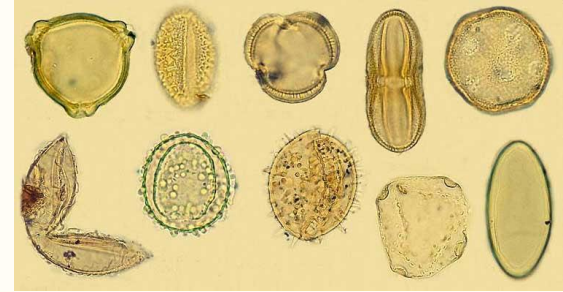
Fire-driven rewiring of ecological networks over 10,000 years

10 000 years of pollen sediments



BSM
Pyrenees, Spain
1913 m

GGU
Bale Mountains,
Ethiopia 3950 m

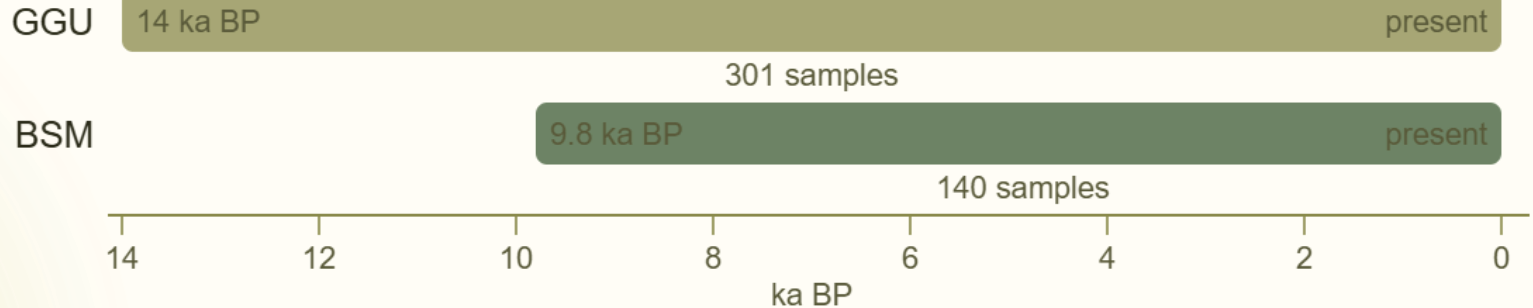
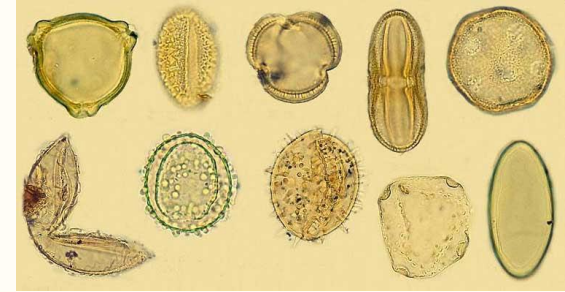


10 000 years of pollen sediments



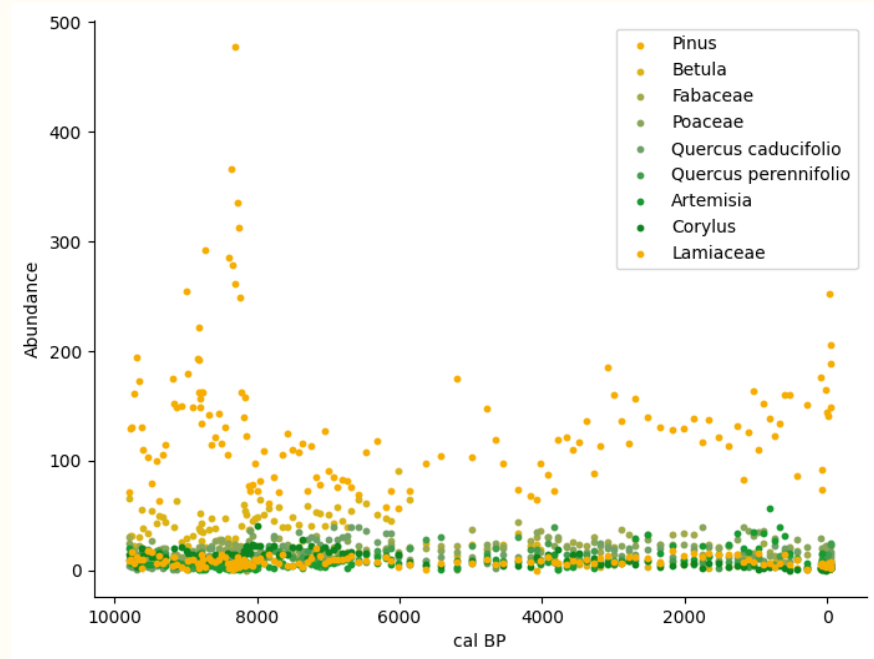
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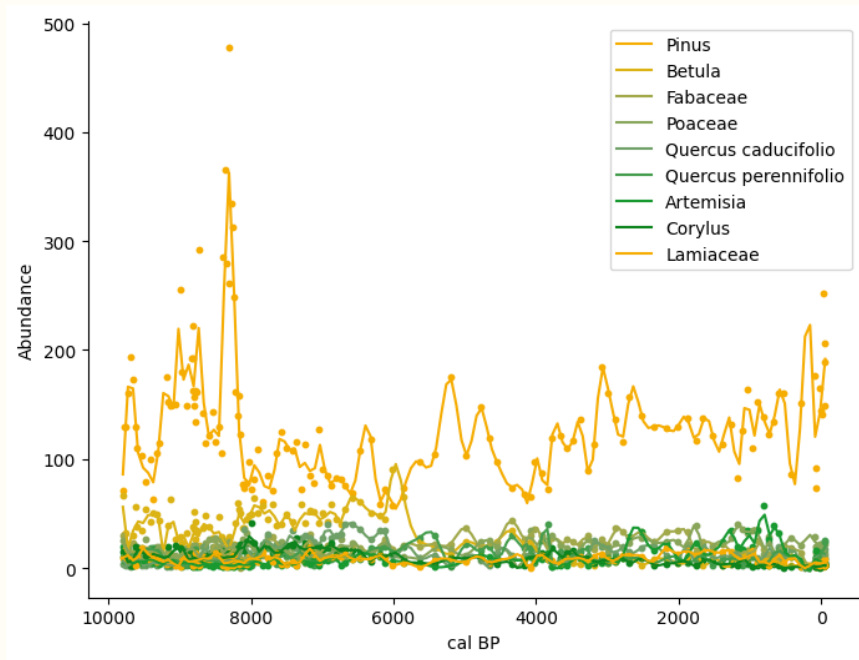
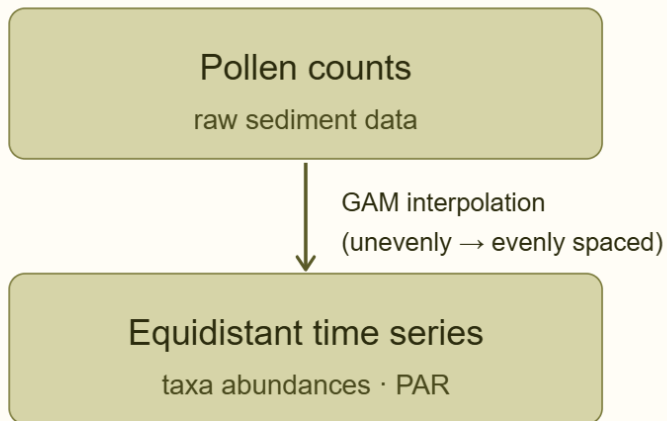
From sediment to causal network

Pollen counts
raw sediment data



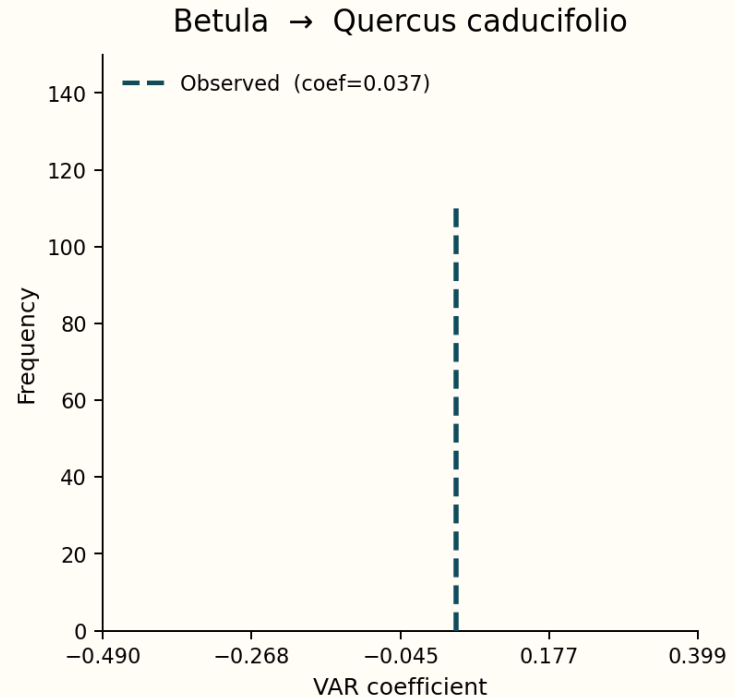
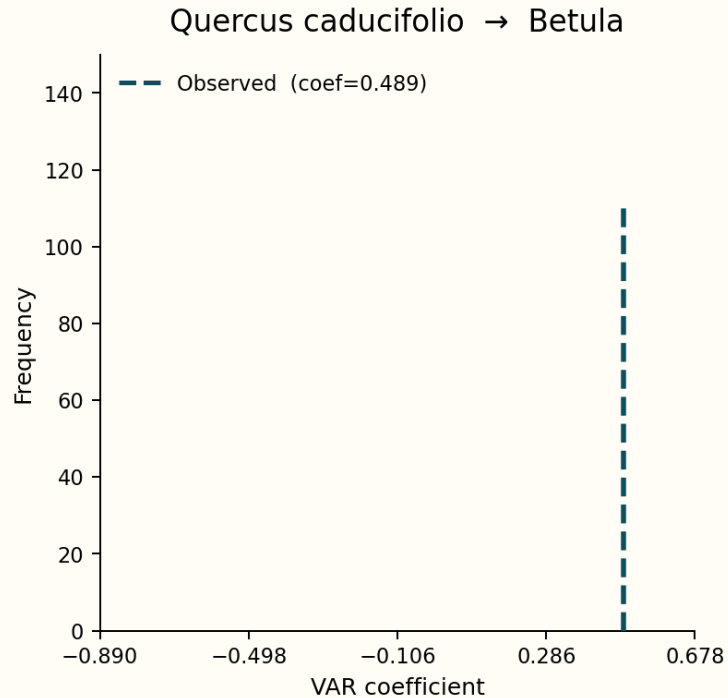
Pollen from the ten most abundant taxa. BSM.

From sediment to causal network



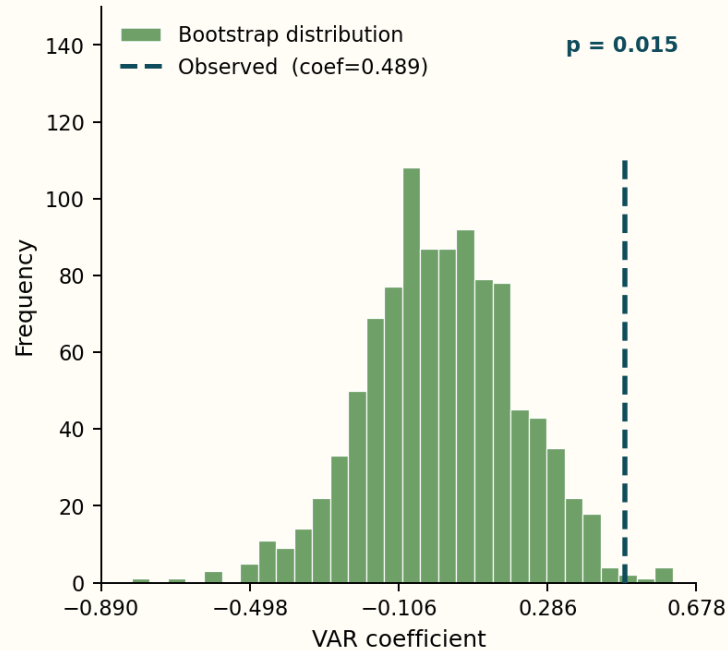
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From sediment to causal network

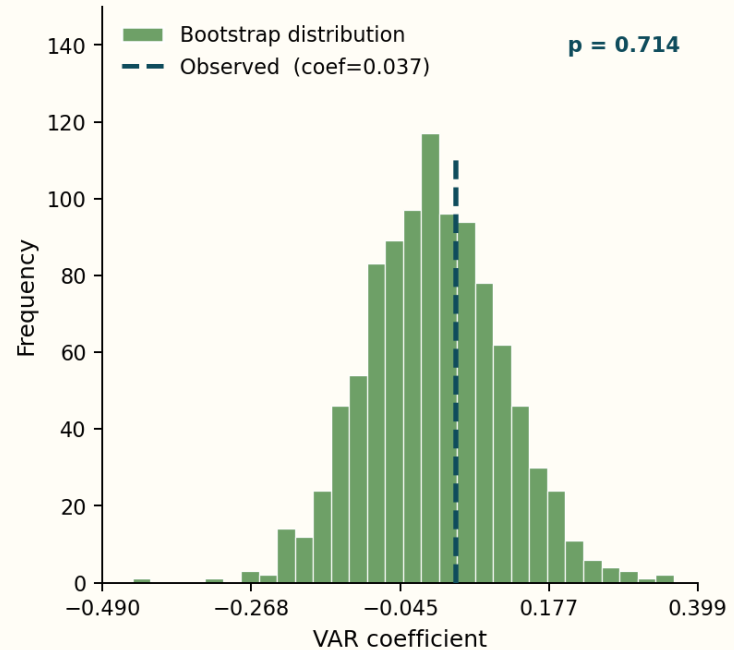


From sediment to causal network

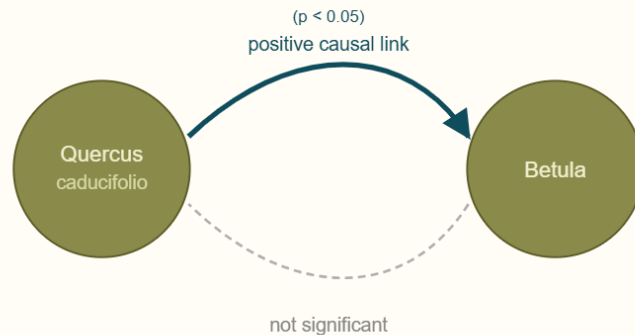
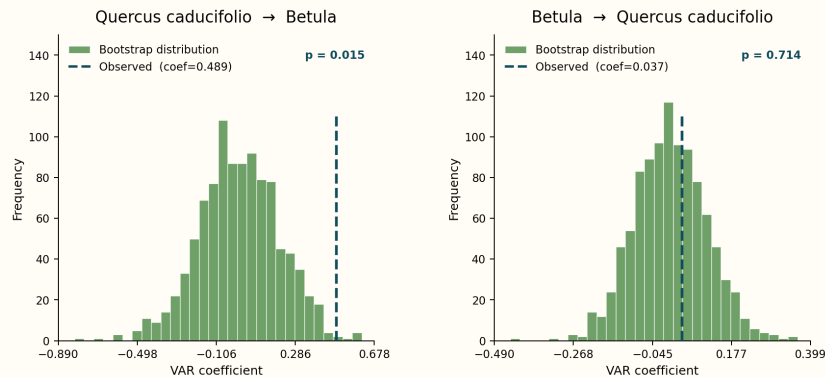
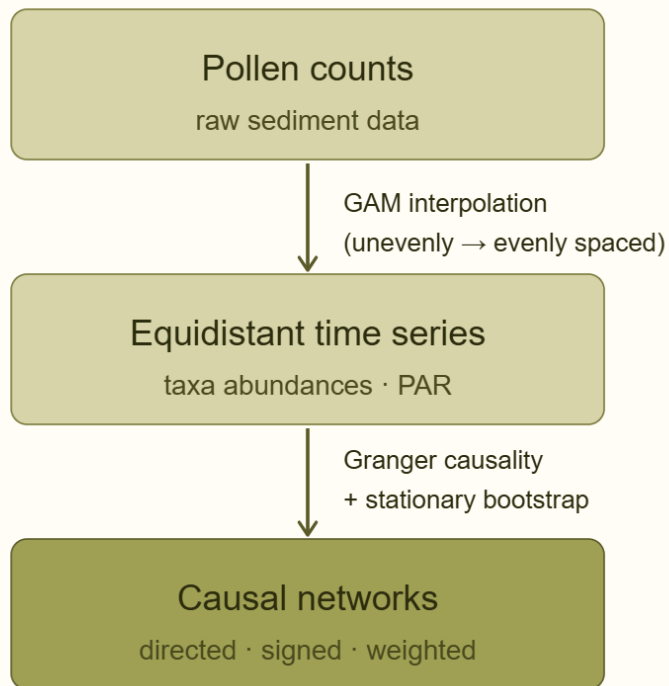
Quercus caducifolio → Betula



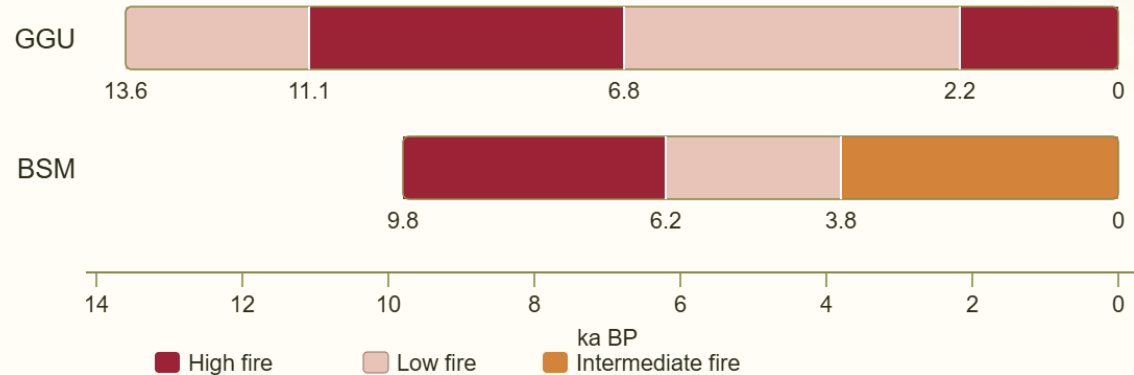
Betula → Quercus caducifolio



From sediment to causal network

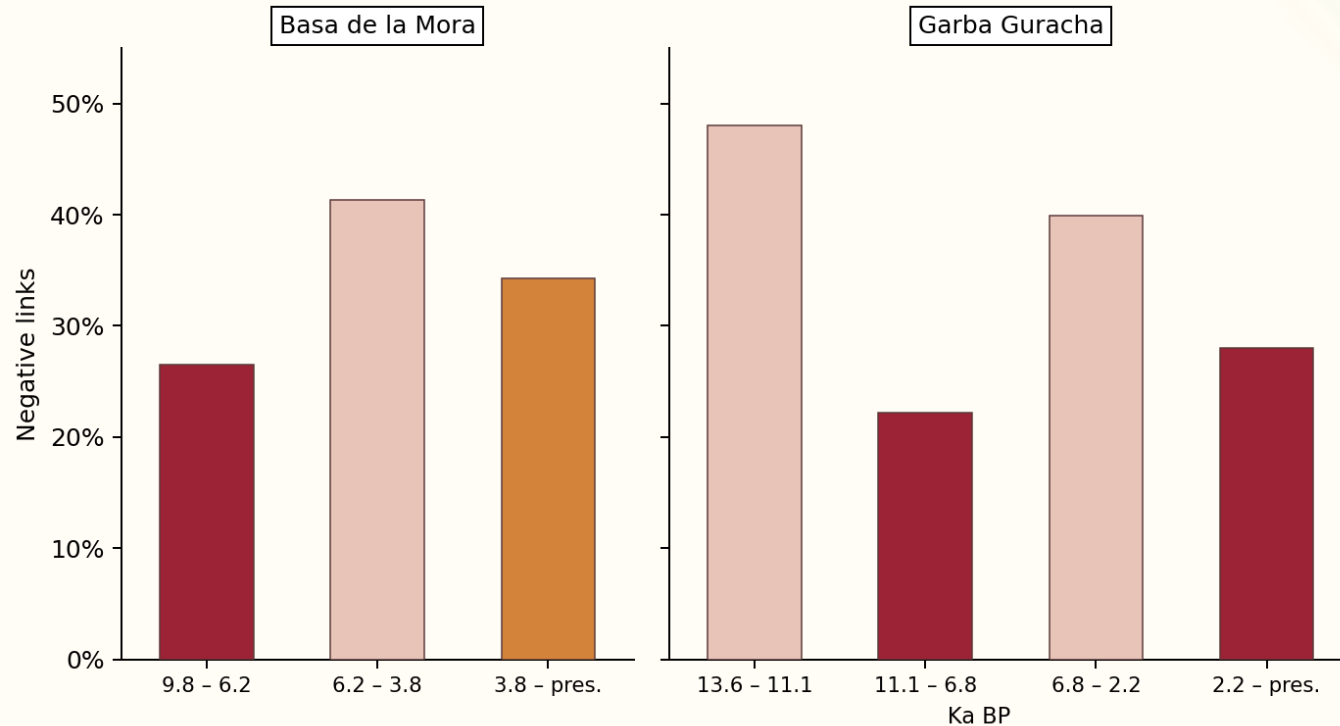


Dividing the record by fire regime

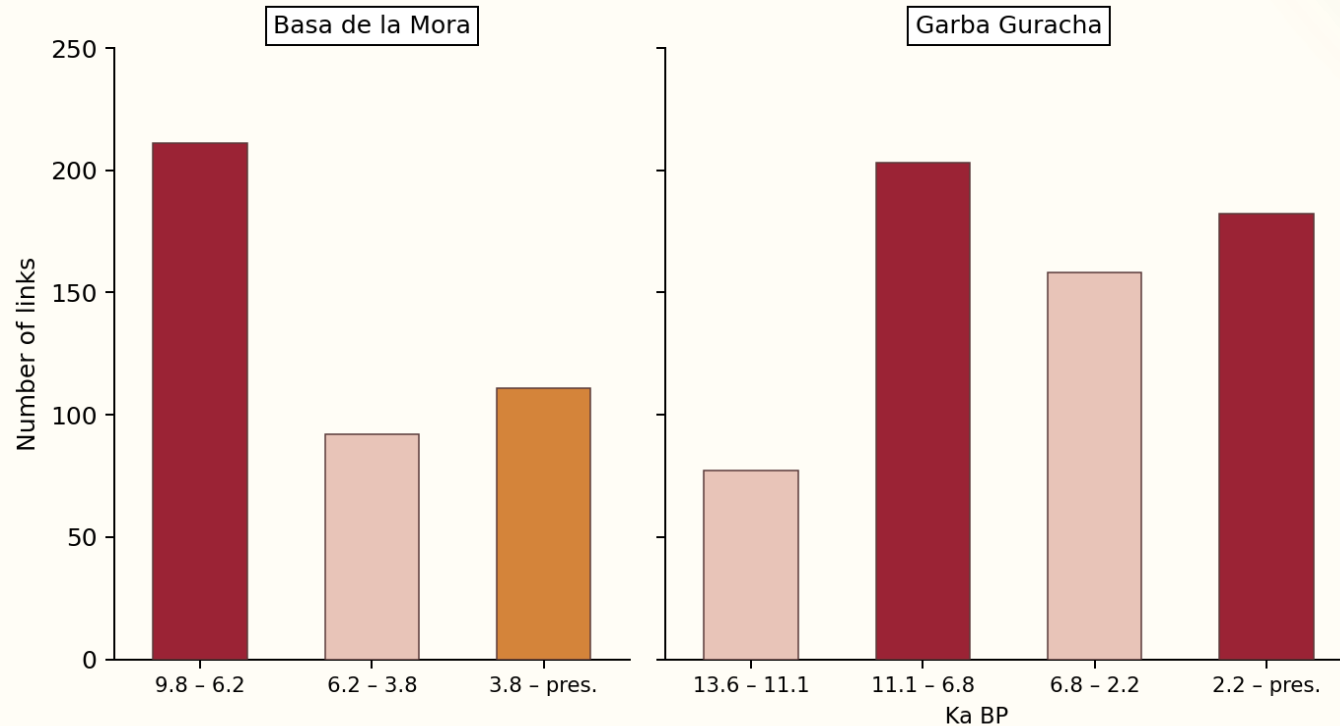


We Split the time series into several adjacent non-overlapping subsets corresponding to distinct regimes of fire activity, quantified according to charcoal concentration.

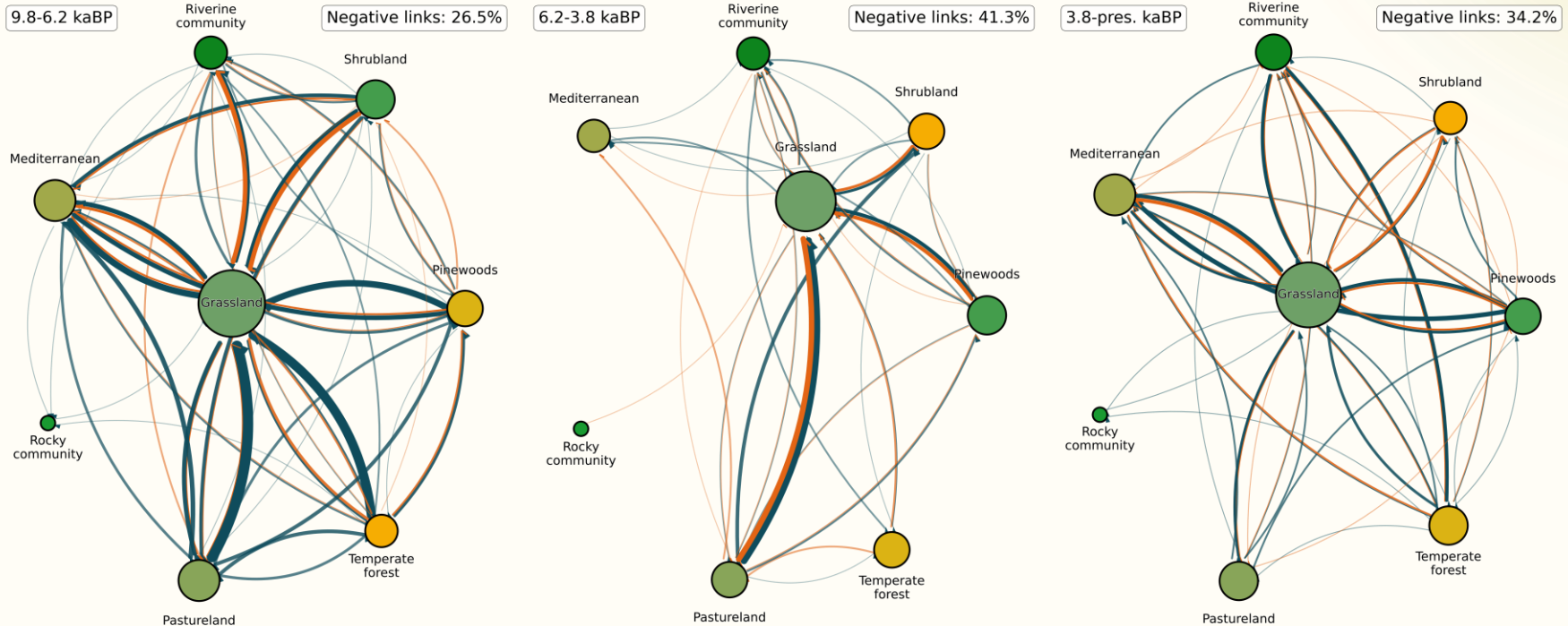
Influence of fire activity on interactions



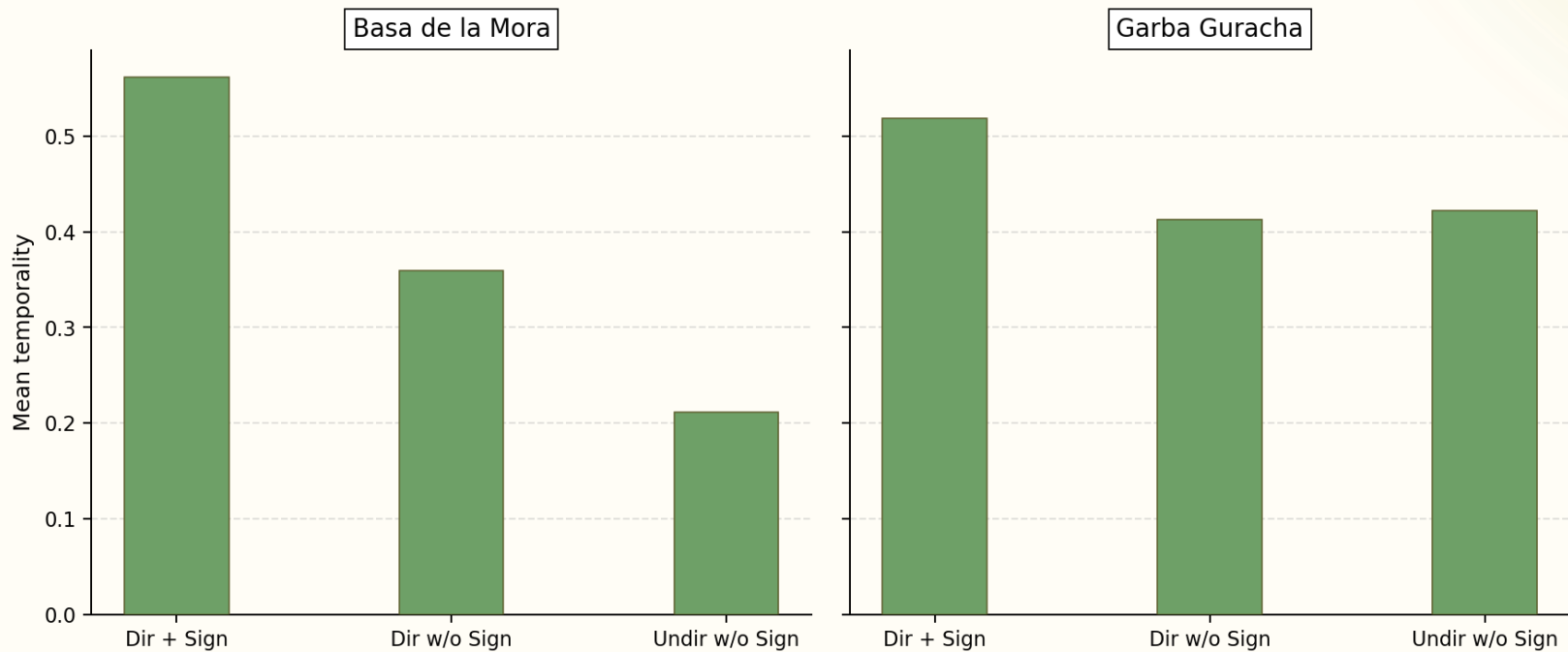
Influence of fire activity on interactions



Fire rewires the network architecture



Fire rewires the network architecture



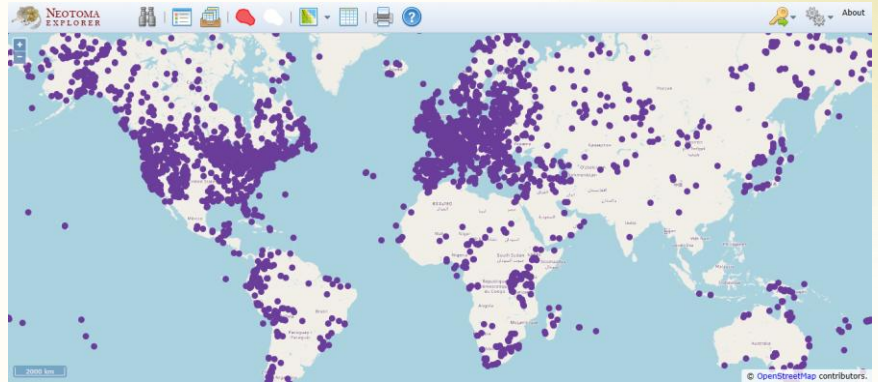
NEXT STEPS

- Beyond link sign: quantifying interaction persistence and rewiring dynamics across functional groups

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Beyond link sign: quantifying interaction persistence and rewiring dynamics across functional groups

Expanding the dataset: replicating the causal network framework on new paleoecological records



Neotoma data base.



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Do you have any
questions?

Thanks!